Emma Dann

Department of Genetics, Stanford University, Palo Alto, California 94305-5120 (USA)
emmadann@stanford.edu | emdann.github.io | Google Scholar

Education

PhD in Biological Sciences (Computational Biology) – University of Cambridge, UK (Oct 2019 – Oct 2023)
Advisors: Prof. Sarah Teichmann, Dr. John Marioni
Thesis title: *Discovering variation from cell atlases: comparative methods for single-cell genomics*

MSc in Cancer, Stem Cells and Developmental Biology (Bioinformatics profile) – Utrecht University, NL (Sep 2017 – Aug 2019)

BSc in Biomolecular Sciences and Technology – University of Trento, IT (Sep 2014 – Jul 2017)

Research experience

Postdoctoral fellow – **Stanford University**, Palo Alto, US (May 2024 – present)
Visiting postdoctoral fellow – **Gladstone-UCSF Institute of Genomic Immunology**, San Francisco, US
Advisors: Prof. Jonathan Pritchard, Prof. Alex Marson

Graduate researcher – **Wellcome Sanger Institute**, Cambridge, UK (Oct 2019 – Jan 2024)
Advisors: Prof. Sarah Teichmann, Dr. John Marioni

Research intern – **European Molecular Biology Laboratory**, Heidelberg, DE (Sep 2018 – Aug 2019)
Advisors: Dr. Wolfgang Huber

Research intern – **KNAW Hubrecht Institute**, Utrecht, NL (Sep 2017 – Aug 2018)
Advisors: Prof. Alexander van Oudenaarden

Research intern – **Center for Integrative Biology**, University of Trento, IT (Feb 2017 – Jul 2017)
Advisors: Prof. Francesca Demichelis

Fellowships & awards

Fellowships

Helen Hay Whitney fellowship (Nov 2024 – present)
Postdoctoral fellowship, awarded $229,500 over 3 years

European Molecular Biology Organization (EMBO) long-term fellowship (Jul 2024 - present)
Postdoctoral fellowship, awarded €130,800 over 2 years

Wellcome Sanger Institute PhD studentship, Wellcome Trust, UK (Oct 2019 - Oct 2023)
PhD fellowship, awarded £100,000 plus tuition over 4 years

U/Select honours programme Utrecht University, NL (Jan 2018 - Oct 2019)
Master's honors program, awarded €2,000 travel grant for international research internship

Merit Award University of Trento, IT (Oct 2017)
Prize for outstanding graduates, awarded €3,000

Funding

Anthropic AI for Science program (2026)
Awarded $20,000 over 6 months

Contributed as key personnel to NIH R01 grant (2R01HG008140) (2025)
PIs: Jonathan K. Pritchard, Alex Marson
Reviewed as 1st percentile score

Contributed as key personnel to ERC Advanced Grant (ThyDESIGN) (2022)
PI: Sarah Teichmann
Awarded €2.5M over 5 years

Other awards

Tahoe Deep Dive Hackathon, first prize winner (May 2025)
Awarded $20,000 for best project on prediction of clinical outcomes from large-scale drug perturbation datasets.

Publications

► denotes manuscripts as leading author, * denotes equal contribution

Peer-reviewed Manuscripts

► Zhu R.*, **Dann E.***, Yan J., Reyes Retana J., Goto R., Guitche R.C., Brixi L., … , Pritchard J.K., Marson A. (2026) Genome-scale perturb-seq in primary human CD4+ T cells maps context-specific regulators of T cell programs and human immune traits. *Cell* | Full text | code | data

Liu Q., Chen P.A., Urs E., … , **Dann E.**, Modak C., … , Marson A., Carnevale J. (2026) *In vivo* genome-wide CRISPR screens in human T cells to enhance T cell therapy for solid tumors. *Nature* | Full text

Jung H., Devant P., Ching C., Ota M., **Dann E.**, Zhu R., … , Marson A., Carnevale J. (2026) Virus-like particles enable targeted gene engineering and pooled CRISPR screening in primary human myeloid cells. *Nature Biotechnology* | Full text

Rathore U., Dugan E., Thornton H., … , **Dann E.**, … , Marson A. (2026) Systematic discovery of pro- and anti-HIV host factors in primary human CD4+ T cells. *Cell* | Full text

Deconinck L., Zappia L., Cannoodt R., Morgan M., **scverse core**, … , Saeys Y. (2025) *anndataR* improves interoperability between R and Python in single-cell transcriptomics. *Bioinformatics* | Full text

Heumos L., Ji Y., … , **Dann E.**, … , Theis F.J. (2025) Pertpy: an end-to-end framework for perturbation analysis. *Nature Methods* | Full text | python package

Ota M., Spence J.P., Zeng T., **Dann E.**, Milind N., Marson A., Pritchard J.K. (2025) Causal modelling of gene effects from regulators to programs to traits. *Nature* | Full text

Luecken M., Gigante S., Burkhardt D., Cannoodt R., … , **Dann E.**, … , Saeys Y., Theis F.J., Krishnaswamy S. (2025) Defining and benchmarking open problems in single-cell analysis. *Nature Biotechnology* | Full text

Schuster V., **Dann E.**, Krogh A., Teichmann S.A. (2024) multiDGD: A versatile deep generative model for multi-omics data. *Nature Communications* | Full text | python package

Yayon N., Kedlian V., Boehme L., … , **Dann E.**, … , Uhlmann V., Notarangelo L.D., Germain R.N., Radtke A.J., Marioni J.C., Taghon T., Teichmann S.A. (2024) A spatial human thymus cell atlas mapped to a continuous tissue axis. *Nature* | Full text

Oliver A.J., Huang N., … , **Dann E.**, …, Elmentaite R., Teichmann S.A. (2024) Single-cell integration reveals metaplasia in inflammatory gut diseases. *Nature* | Full text

Missarova A., **Dann E.**, … , Marioni J.C. (2023) Leveraging neighborhood representations of single-cell data to achieve sensitive DE testing with miloDE. *Genome Biology* | Full text | R package

Sumanaweera D., Suo C., Muraro D., **Dann E.**, … , Teichmann S.A. (2023) Gene-level alignment of single cell trajectories. *Nature Methods* | Full text | python package

Barnes J. L., He P., … , **Dann E.**, … , Teichmann S.A., Meyer K. B., Nikolic M. Z. (2023) Early human lung immune cell development and its role in epithelial cell fate. *Science Immunology* | Full text

► **Dann E.**, Cujba A.M., Oliver A., Meyer K., Teichmann S.A., Marioni J.C. (2023) Precise identification of cell states altered in disease using healthy single-cell references, *Nature Genetics* | Full text | code | python package

Botting R.A., Goh I., … , **Dann E.**, … , Teichmann S.A., Haniffa M. (2023) Yolk sac cell atlas reveals multiorgan functions during human early development. *Science* | Full text

Suo C., Polanski K., **Dann E.**, … , Tuong Z.K., Clatworthy M., Teichmann S.A. (2023) Dandelion uses the single-cell adaptive immune receptor repertoire to explore lymphocyte developmental origins, *Nature Biotechnology* | Full text | python package

Heumos L., … **Single-cell Best Practices Consortium**, … , Theis F.J. (2023) Best practices for single-cell analysis across modalities. *Nature Reviews Genetics* | Full text

► Suo C.*, **Dann E.***, … , Haniffa M., Teichmann S.A. (2022) Mapping the developing human immune system across organs. *Science* | Full text | code | data

► Lalchand V.*, Ravuri A.*, **Dann E.***, …, Madad S., Teichmann S.A., Lawrence N.D. (2022) Modelling Technical and Biological Effects in scRNA-seq data with Scalable GPLVMs. *Proceedings of the 17th Machine Learning in Computational Biology meeting* | Full text

He P., Lim K., Sun D., … , **Dann E.**, … , Meyer K.B., Rawlins E.L. (2022) A human fetal lung cell atlas uncovers proximal-distal gradients of differentiation and key regulators of epithelial fates, *Cell* | Full text

Kleshchevnikov V., Shmatko A., **Dann E.**, …, Stegle O., Bayraktar O.A. (2022) Cell2location maps fine-grained cell types in spatial transcriptomics. *Nature Biotechnology* | Full text | python package

► **Dann E.**, Henderson N.C., Teichmann S.A., Morgan M.D., Marioni J.C. (2022) Differential abundance testing on single-cell data using k-nearest neighbor graphs, *Nature Biotechnology* | Full text | R package | python package

Jardine L., Webb S., … , **Dann E.**, … , Göttgens B., Roberts I., Teichmann S.A., Haniffa M. (2021) Blood and immune development in human fetal bone marrow and Down syndrome, *Nature* | Full text

Elmentaite R., Kumasaka N., Roberts K., Fleming A., **Dann E.**, … , James K.R., Teichmann S.A. (2021), Cells of the human intestinal tract mapped across space and time, *Nature* | Full text

Stephenson E., Reynolds G., Botting R.A., Calero-Nieto F.J., Morgan M.D., Tuong Z.K., Bach K., Sungnak W., … , **Dann E.**, … , Duncan C.J.A., Smith K., Teichmann S.A., Clatworthy M.R., Marioni J.C., Göttgens B., Haniffa M. (2020) Single-cell multi-omics analysis of the immune response in COVID-19, *Nature Medicine* | Full text

Beltran H., Romanel A., … , **Dann E.**, … , Benelli M., and Demichelis F. (2020) Circulating Tumor DNA Profile Recognizes Transformation to Castration-Resistant Neuroendocrine Prostate Cancer, *Journal of Clinical Investigation* | Full text

Pre-print Manuscripts

► **Dann E.***, Spence J.P.*, Ota M., Engreitz J., Marson A., Pritchard J.K. (2026) Trait Causal Models: from GWAS, perturbations, and omics to predictive models of human biology. *Zenodo (preprint)* | Full text

Hartman A., Blair J., … , **Dann E.**, … , Satpathy A., Roth T. (2026) Systematic identification of seed-driven off-target effects in Perturb-seq experiments. *bioRxiv* | Full text

Pett J.P., Prete M., Pham D., England N., … , **Dann E.**, … , Teichmann S.A., Meyer K. (2025) An atlas of TF driven gene programs across human cells. *bioRxiv* | Full text

► **Dann E.***, Teeple E.*, Elmentaite R., … , Rinaldis E., Teichmann S.A. (2024) Estimating the impact of single-cell RNA sequencing of human tissues on drug target validation. *medRxiv* | Full text | code

Presentations

Conference oral presentations

June 2026 - Biology at Scale conference, Wellcome Genome Campus, Hinxton (UK)
May 2026 - Warren Alpert Computational Biology and AI Research Symposium, Stanford (US)
Feb 2026 - Arc Institute Symposium, San Francisco (US)
Apr 2025 - Chan Zuckerberg Initiative Cell Science Meeting, Chicago (US)

Nov 2022 - Machine Learning in Computational Biology conference, virtual
Jun 2022 - Single Cell Biology conference, Wellcome Genome Campus, Hinxton (UK)
May 2022 - The Biology of Genomes, Cold Spring Harbor Laboratory (US)
Oct 2021 - Chan Zuckerberg Initiative Single Cell Biology Meeting, virtual
Sep 2021 - Next Generation Genomics Meeting, virtual
Mar 2021 - CZI Seed Networks Computational biology meeting, virtual

Invited research seminars

Apr 2026 - Emerging Genomic Scientists Symposium, University of Pennsylvania, Philadelphia (US)
Oct 2025 - Ultima Genomics user group session @ American Society for Human Genetics, Boston (US)
Sep 2025 - scverse community meeting, virtual
Jul 2025 - AI + Biomedicine seminar, Stanford (US)
Sep 2024 - BRAID Genentech, South San Francisco (US)
Sep 2024 - Human Genetics department, University of Chicago, Chicago (US)
Sep 2024 - SQLIFTS Lecture Series, Northwestern University, Chicago (US)
Mar 2024 - Human Technopole, Milan (IT)
Dec 2023 - Google DeepMind, London (UK)
Jul 2023 - Sanofi Precision Medicine & Computational biology, Cambridge (US)
Jun 2023 - Chan Zuckerberg Initiative, Redwood City (US)
Apr 2023 - Cambridge AI club for biomedicine, Milner Institute, Cambridge (UK)
Dec 2022 - UMC Utrecht single-cell genomics meeting, virtual
Sep 2022 - UK Conference of Bioinformatics and Computational Biology, virtual
Jul 2022 - Computational Health Center seminar, Helmholtz Munich (DE)
Apr 2022 - Maxwell Society Annual Conference, King's College London (UK)
Apr 2022 - NYU Langone single cell journal club, virtual
Mar 2021 - Cambridge Center for Physical Biology Single Cell Symposium, virtual

Guest lectures

Aug 2024 - Why there's never been a better time to study human biology, Girls in Research program (virtual)
Aug 2023 - Intro to single-cell genomics series, University of Sri Jayewardenepura (virtual)
Mar 2023 - Introduction to multiomics data integration and visualisation, EMBL-EBI training course (UK)
Apr 2022 - Best practices and open challenges in single-cell genomics data analysis, Maxwell Society Annual Conference, King's College London (UK)

Conference poster presentations

Oct 2025 - American Society for Human Genetics conference, Boston (US)
Jul 2023 - Human Cell Atlas General Meeting, Toronto (CA)
Oct 2022 - Single Cell Genomics conference, Utrecht (NL)
Jun 2021 - Human Cell Atlas General Meeting, virtual

Teaching & mentoring

Computational research assistant recruitment and mentoring, Marson lab (Fall 2025 - present)

PhD rotation student recruitment and mentoring, Pritchard lab (Winter 2025 - present)

Stanford & Tanzania Human Genetics Organization virtual training program (Spring 2025)
One-on-one virtual mentoring program on research project for students from Tanzania

PhD rotation student recruitment and mentoring, Teichmann lab (Fall 2021 - Fall 2023)

Systems biology: From large datasets to biological insight, EMBL-EBI training course (Jul 2022)
Invited instructor for single-cell multi-omics module (course material)

Advanced topics in Single Cell Genomics, Swiss Institute for Bioinformatics course (Apr 2022)
Invited instructor for multi-omics module (course material)

Advanced topics in Single Cell Genomics, SciLifeLab / Swiss Institute for Bioinformatics Summer School (Aug 2021)
Invited instructor for multi-omics module (course material)

Leadership & outreach

Core team member, scverse consortium (Feb 2024 – present)
I am part of the governance team of the consortium, overseeing the development and maintenance of foundational software tools for omics data analysis.

- Organizational governance and strategic planning
- Development of academic and industry partnerships for funding initiatives and collaborative research projects
- Lead organizer of the scverse annual conference and community hackathons
- Contributor to open-source tutorials and resources for newcomers to single-cell genomics analysis in python

Competition judge, Open Problems Single-cell perturbations Kaggle competition (Fall 2023)
Served as judge for NeurIPS-affiliated machine learning challenge focused on predicting drug perturbation responses.

Co-organizer, Strengthening integration pipelines session, Human Cell Atlas general meeting (Jul 2023)
Co-organized and led panel discussion on cell atlas construction and usage.

Member of the selection panel, Sanger prize competition (Mar 2023)
The Sanger Prize competition funds research internships at Sanger for undergraduate students from LMI countries. I was a member of the Grad Panel, evaluating essays from applicants.

Co-organizer Open Problems in Single-Cell Analysis Jamboree (April 2021)
Led task for benchmarking of compositional analysis methods for scRNA-seq.

Co-organizer, EBI-Sanger-Cambridge PhD Symposium (Feb 2020)
I was in charge of the scientific program and sponsorship for the PhD program symposium.

Board member, Open Wet Lab (OWL) (Oct 2015 – Jan 2018)
OWL is the first biohacking organization in Italy. I was in charge of the association's budget and directed the publication of OWL's blog and scientific outreach articles for local newspapers.

Peer review

Journals: Nature (1 paper), Cell (1* paper), Nature Biotechnology (2* papers), Nature Genetics (1 paper), Nature Communications (1 paper), Genome Biology (1 paper), Bioinformatics Advances (1 paper), OUP Bioinformatics (1 paper), Reviewer Commons (1 paper)
*Assisting a senior reviewer

Conferences: Machine Learning in Computational Biology (2023, 2024, 2026)

Grants/fellowships: Chan Zuckerberg Initiative Single Cell Biology Data Insights, Christian Doppler Research Association Transfer.S2S grants

Software

pertpy | scverse/pertpy | contributor | perturbation analysis toolkit for the scverse ecosystem.

miloR | MarioniLab/miloR | primary developer | R/Bioconductor package for the identification of differential cell abundance in KNN graphs for single-cell data

milopy | emdann/milopy | primary developer | python package for the identification of differential cell abundance in KNN graphs for single-cell data

genomic-features | scverse/genomic-features | primary developer | Genomic annotations using Bioconductor resources in Python.

oor_benchmark | MarioniLab/oor_benchmark | primary developer | python framework to benchmark workflows for detection of out-of-reference cells in comparative single-cell analysis

cell2location | BayraktarLab/cell2location | contributor | python package for mapping of tissue cell types via integration of single cell RNA-seq and spatial transcriptomics data

Industry experience

Data science consultant – Ensocell Ltd (Oct 2023 – Mar 2024)
Target discovery platform and analysis of single-cell data